Coronary Heart Disease (CHD) Prediction

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Project Context

Objective and Reasoning :

Coronary Heart Disease (CHD) is one of the leading causes of death globally. Early prediction can help prevent severe outcomes.

Build a machine learning model to predict the 10-year CHD risk using the Framingham Heart Study dataset from Kaggle.

Tools:

-PySpark for scalable data processing and machine learning.

-Matplotlib and Seaborn for visualization.

Methodology

1. Data Preprocessing

-Loaded the Framingham dataset.

-Dropped irrelevant column: education.

-Renamed male to Sex_male for clarity.

-Cast columns cigsPerDay, totChol, and glucose to DoubleType for compatibility with ML algorithms.

-Removed rows with missing values. (NULL values)

```
# Load dataset
file_path = "framingham.csv"
disease_df = spark.read.csv(file_path, header=True, inferSchema=True) # header=Tru
```

```
# Drop the 'education' column
disease_df = disease_df.drop('education')
```

```
# Rename column 'male' to 'Sex_male'
disease_df = disease_df.withColumnRenamed('male', 'Sex_male')
```

```
# Check for string-type columns and cast to DoubleType
columns_to_cast = ['cigsPerDay', 'totChol', 'glucose']
for col_name in columns_to_cast:
    disease_df = disease_df.withColumn(col_name, col(col_name).cast(DoubleType()))
```

```
# Remove rows with null values
disease_df = disease_df.na.drop()
```

Methodology

2. Feature Engineering

-Selected key features:

age, Sex_male, cigsPerDay, totChol, sysBP, glucose.

```
# Features and target variable
feature_columns = ['age', 'Sex_male', 'cigsPerDay', 'totChol', 'sysBP', 'glucose']
assembler = VectorAssembler(inputCols=feature_columns, outputCol="features")
disease_df = assembler.transform(disease_df)

#combines multiple feature columns into a single vector thats relevant to the model
```

-Combined features into a single vector using VectorAssembler.

-Standardized features using StandardScaler to improve model performance.

```
# Standardize features
scaler = StandardScaler(inputCol="features", outputCol="scaledFeatures", withStd=True, withMean=False)
scaler_model = scaler.fit(disease_df)
disease_df = scaler_model.transform(disease_df)
# this step improves the performance of the model by standerdizing the feature values to have unit variance
```

Methodology

3. Model Training

-Split data into training (70%) and testing (30%).

-Trained a Logistic Regression model using scaled features.

```
# Select features and target for modeling
data = disease_df.select('scaledFeatures', 'TenYearCHD')
```

```
# Train-Test Split
train_data, test_data = data.randomSplit([0.7, 0.3], seed=42) #for evaluating model performance
```

```
# Logistic Regression Model
lr = LogisticRegression(featuresCol='scaledFeatures', labelCol='TenYearCHD')
lr_model = lr.fit(train_data)

# logistic Regression is a classification algorithm used to predict categorical data (binary classification)
```

Results

1. Model Accuracy

-The model achieved an accuracy of 85%

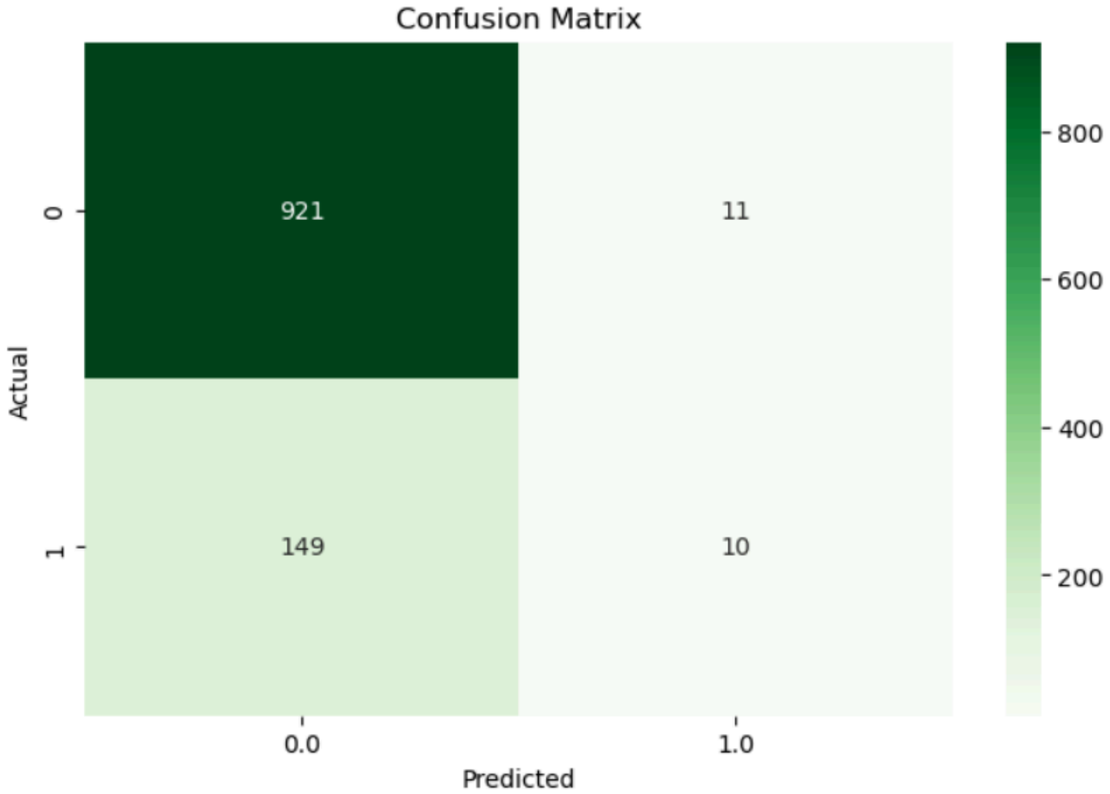
```
# Evaluation
evaluator = MulticlassClassificationEvaluator(
    labelCol="TenYearCHD", predictionCol="prediction", metricName="accuracy"
)
accuracy = evaluator.evaluate(predictions)
print(f"Accuracy of the model is = {accuracy:.2f}")

#Evaluates the model accuracy using MCE which measures the proportion of correctly classified instances
```

```
Accuracy of the model is = 0.85
```

-Which means for example, out of 100 instances 85 were classified correctly

Results

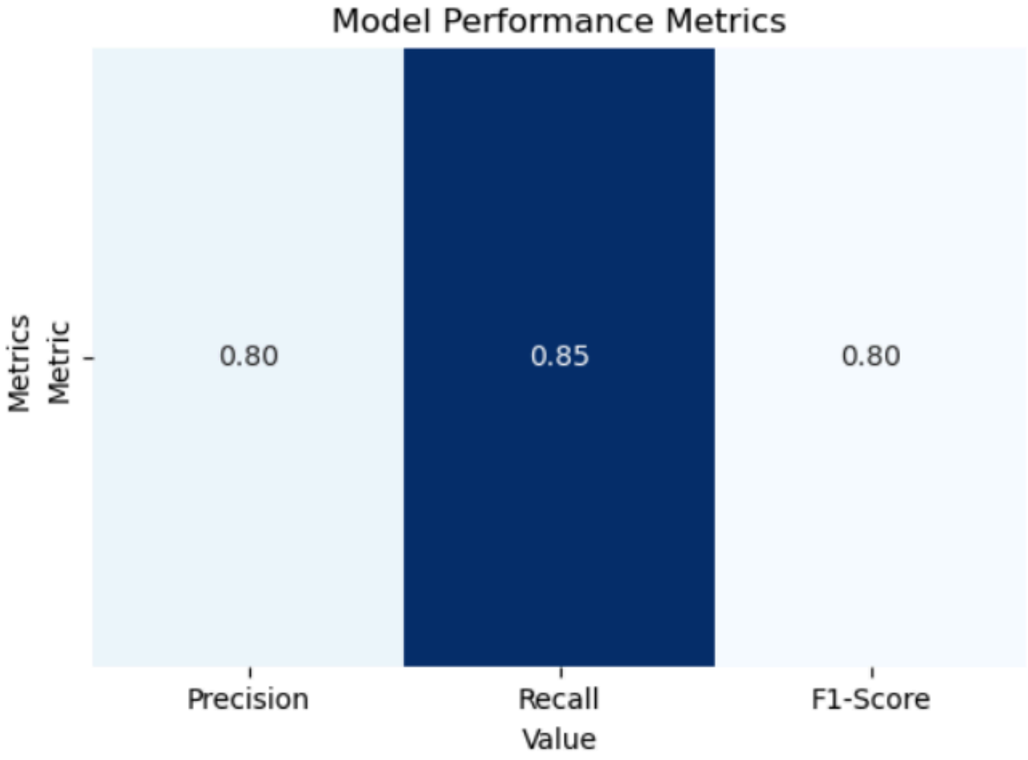
2. Confusion Matrix



-**True Positives (TP): Correctly predicted CHD cases. (actual 1.0 predicted 1.0)**
-**True Negatives (TN): Correctly predicted non-CHD cases.(actual 0.0 predicted 0.0)**
-**False Positives (FP): Predicted CHD incorrectly. (actual 0.0 predicted 1.0)**
-**False Negatives (FN): Missed CHD cases. (actual 1.0 predicted 0.0)**

Results

3. Performance Metrics



Precision: Proportion of predicted CHD cases that are correct.
Recall: Proportion of actual CHD cases detected.
F1-Score: Balance between precision and recall.

Conclusion and Future Improvements

Conclusion:

-The model successfully predicts CHD risk with reasonable accuracy.
-Logistic Regression is effective for binary classification tasks like this one.

Future Improvements:

-Incorporate additional features or datasets to improve prediction.
-Experiment with other machine learning models like Random Forest or Gradient Boosting.
-Optimize hyperparameters to improve performance.

Thank you for your Attention !